

The Application of Evolutionary Computation in Evacuation Planning

Rani Muhdi, Aaron Garrett, Ravikant Agarwal, Jerry Davis, Gerry Dozier, and David Umphress

Abstract—Evacuation planning plays a significant role in building evacuation. The purpose of this paper is to demonstrate how an evolutionary computation technique in the form of an Estimation of Distribution Algorithm can be used in evacuation planning. This technique is used to evolve the number and location of exits in order to minimize overall evacuation time and reduce the number of casualties and injuries. The algorithm is applied to three day-care layouts, classified as playroom, lunchroom, and classroom settings. The algorithm generates an optimal or near-optimal configuration, and results across several trials can be used to determine the probability that an exit is needed for each possible location. The best exit configurations are presented for each of the three layouts, and a brief analysis is discussed. Although evacuation planning presented in this paper is focused on room layouts, evolutionary computation techniques have the potential to be implemented in large-scale evacuation planning.

I. INTRODUCTION

Evacuation procedures and planning continue to present a significant challenge to building occupants and emergency responders during evacuation and rescue operations. Part of the challenge is caused by the creative designs and intricate structures of many buildings, which can usually be overcome by practicing emergency evacuation drills. Although real emergencies and experimental evacuation drills have some similarities between them [1], performing an evacuation drill still introduces ethical, practical, and financial issues to emergency planners [2]. Therefore, evacuation models offer a potential alternative to overcome these challenges, and confidently assure safety in buildings. The development of computer-based evacuation models in the last three decades has contributed to predicting some of the key elements associated with evacuation. Some of these key elements include evacuation time, number of fatalities, occupant movement, and human behavior. However, the stochastic nature of evacuations, created mainly by the human element and the hazard(s) present, makes it more difficult to achieve a realistic level of modeling.

Another contribution of evacuation models is that of assessing the safety of buildings designs, and compliance with the Life Safety Code®. Some advanced evacuation models rely on computer-aided design (CAD) drawings to generate building layouts before an evacuation is simulated. The simulation outputs are directly dependent on building

layouts. The Life Safety Code® [3] defines means of egress as “a continuous and unobstructed way of travel from any point in a building or structure to a public way consisting of three separate and distinct parts: (1) the exit access, (2) the exit, and (3) the exit discharge.” In addition, the geometry of a building, the location of exits, and the number of exits dictate the means of egress for all those occupying a building. The purpose of this paper is to introduce how an evolutionary computation technique, namely Estimation of Distribution Algorithm (EDA), can be used in evacuation planning by evolving the location of exits in order to optimize (minimize) overall evacuation time.

II. EVOLUTIONARY COMPUTATION

A. Introduction to Evolutionary Computation (EC)

Evolutionary computation techniques [4]–[10], such as genetic algorithms [4] and evolutionary programming [9], are robust optimizers that have been successfully applied to a wide range of problems. These techniques apply the same search principle, natural selection, in an effort to evolve optimal or near-optimal solutions.

Fig. 1 provides an example of the evolutionary process of all ECs. Initially, a population of candidate solutions (also referred to as individuals) is randomly generated, and each individual of the population is evaluated using an evaluation function. The evaluation function assigns an individual a fitness value that is representative of how “good” it is. After the initial population has been created and evaluated, the EC iteratively refines the population by: (1) selecting parents based on their assigned fitness, (2) allowing the selected parents to create offspring through crossover (sexual reproduction) and/or mutation (asexual reproduction), (3) evaluating the offspring, and (4) determining which individuals of the current population and the set of offspring survive to the next generation.

```
Procedure EC {  
    t = 0;  
    Initialize P(t);  
    Evaluate P(t);  
    While (Not Done) {  
        Parents(t) = SelectParents(P(t));  
        Offspring(t) = Procreate(Parents(t));  
        Evaluate(Offspring(t));  
        P(t+1) = SelectSurvivors(P(t), Offspring(t));  
        t = t + 1;  
    }  
}
```

Fig. 1. Pseudocode Version of an EC

Rani Muhdi and Jerry Davis, PhD, CPE, CSP, are with the Department of Industrial & Systems Engineering, Auburn University, Auburn, AL 36849, USA

Aaron Garrett and Gerry Dozier, PhD, (ACI Lab), along with Ravikant Agarwal and David Umphress, PhD, are with the Department of Computer Science and Software Engineering, Auburn University, Auburn, AL 36849, USA

B. Estimation of Distribution Algorithms

Estimation of Distribution Algorithms (EDAs) [6] attempt to leverage the statistical properties of the fitness landscape in order to create offspring. In EDAs, there are neither crossover nor mutation operators. During each generation, the new population is created by sampling the probability distribution of the current population.

For binary-coded chromosomes, the EDA makes use of the probability distribution function of selected individuals. For real-coded chromosomes, the probability density function is used instead [6]. In both cases, a set of parents is selected from the population. The probability distribution/density function is calculated for the set, and used to create a new population of offspring. An additional modification to EDAs called *elitism* [6] allows the best K individuals from the current generation to survive into the next generation, regardless of whether they are worse than an offspring. Typically, K is taken to be 1 if not otherwise specified.

EDAs operate along one dimension at a time when creating offspring. Essentially, for binary-coded chromosomes along a particular dimension, a random parent is selected and its gene value is used to create the offspring's gene. For the real-coded chromosomes along a particular dimension, the mean and standard deviation of the set of the parent genes are calculated. Each offspring's gene (along that dimension) is created according to the following equation,

$$offspring_{i,dim} = mean_{dim} + stdev_{dim} \cdot N(0,1) \quad (1)$$

where i is the offspring number, dim represents the dimension, and $N(0,1)$ is a standardized Gaussian random variate, which is different for each offspring.

III. PROBLEM AND REPRESENTATION

According to the Life Safety Code[®], the geometry of a building, the location of exits, and the number of exits dictate the means of egress for all people occupying a building [3]. In this work, we concern ourselves with finding the optimal placement of exits in a specified room in order to evacuate people as quickly as possible.

We chose three different room layouts for our experiments. All three layouts involve a room with dimensions 39- by 22-feet. The first layout represents a day-care playroom containing 57 randomly positioned children and no obstacles. The second represents a day-care lunchroom (i.e., cafeteria) containing 24 children and 7 fixed obstacles, illustrated in Fig. 2. The final layout represents a day-care classroom containing 1 adult, 14 children, and 8 fixed obstacles, illustrated in Fig. 3. In all layouts, children and adults were represented as circles with 1.5-foot and 2-foot diameters, respectively. The area of the room, number of children, and occupant load factor are in compliance with the Life Safety Code[®].

The possible exit locations were determined by calculating how many 3-foot-wide doors would fit along each exterior wall with no separation between them. If a wall did not allow an integral number of doors (as is often the case), the set of doors was centered along the wall with the excess space placed equally in each corner. This process was performed

for each wall. (Therefore, theoretically, each of the three layouts could contain a maximum of 40 exits.) The doors were then numbered starting with the far left door along the north wall and continuing clockwise. In this way, we were able to represent the presence or absence of the exits using a one-dimensional Boolean array. Though this approach limits us to a set of fixed locations for the doors (i.e., there is no way to place a door in an arbitrary location, even if such a placement would lead to a better solution), it allows us to use a fixed-length encoding scheme to represent a variable number of doors. Additionally, it prevents occurrences of invalid chromosomes, such as a chromosome that places a door in such a way that it overlaps another door.

IV. EVACUATION MODEL

In order to assess the fitness of each configuration of exit locations, a model of evacuee escape behavior is needed. The movement of the evacuees was simulated by choosing an artificial potential field model similar to those used in mobile robotics path planning [11]. We chose to use the potential field model for this preliminary work because of its power and simplicity. The model is capable of producing relatively realistic evacuee behavior without requiring any parameter specification.

In our model, each entity (i.e., evacuees, obstacles, and exits) are modeled as elements in a force field. For a given evacuee, the exit locations produce attractive forces, while the obstacles and other evacuees generate repulsive forces. The resultant forces acting on each evacuee form a potential field. Each evacuee then follows the gradient of this field at every time-step of the simulation. Due to this gradient descent approach to motion planning, evacuees may become trapped in local minima of the field. However, this type of indecision on the part of the evacuees may not be entirely unrealistic, and our optimization should attempt to eliminate configurations that produce lots of local minima in the potential field.

In order to calculate the movement of the evacuees, we allow each force acting on a given evacuee to be inversely proportional to the square of the distance between the evacuee and the source (with all proportionality constants

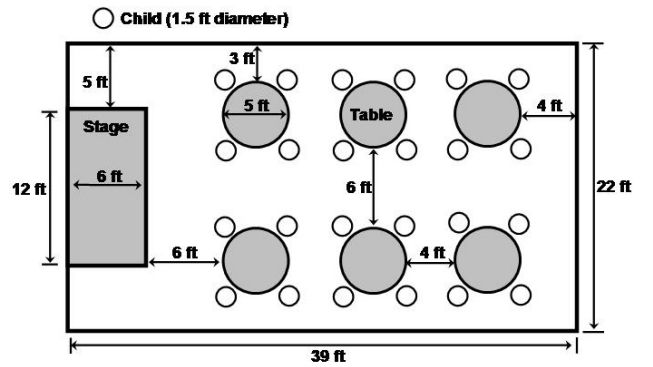


Fig. 2. Day-Care Lunchroom Layout

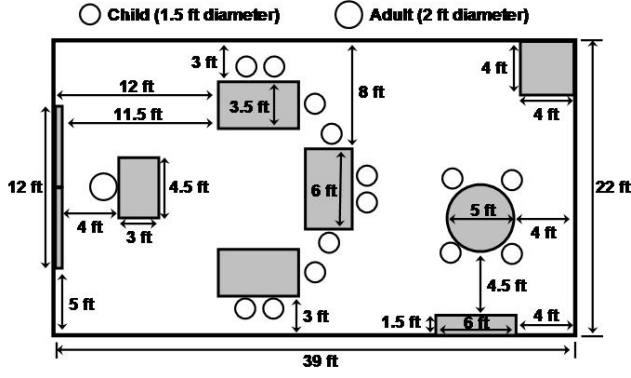


Fig. 3. Day-Care Classroom Layout

set to 1.0 for simplicity). For a given evacuee position $\vec{p}$, the resultant force acting on this evacuee is $F(\vec{p})$, as given in Equation 2.

- $\vec{p}$: The current position of the evacuee
- A : The set of all attractive vectors (exit locations)
- $\vec{a}$: A particular element of A
- O : The set of all obstacle vectors
- $\vec{o}$: A particular element of O
- E : The set of all evacuee vectors
- $\vec{e}$: A particular element of E
- $d(\vec{x}, \vec{y})$: The Euclidean distance between $\vec{x}$ and $\vec{y}$
- $F(\vec{p})$: The resultant force acting on location $\vec{p}$

$$F(\vec{p}) = \sum_{\vec{a} \in A} \frac{1}{d(\vec{p}, \vec{a})^2} - \sum_{\vec{o} \in O} \frac{1}{d(\vec{p}, \vec{o})^2} - \sum_{\substack{\vec{e} \in E \\ \vec{e} \neq \vec{p}}} \frac{1}{d(\vec{p}, \vec{e})^2} \quad (2)$$

The gradient of F is calculated according to Equations 3, 4, and 5.

$$\nabla F = \left\langle \frac{\partial F}{\partial x}, \frac{\partial F}{\partial y} \right\rangle \quad (3)$$

$$\begin{aligned} \frac{\partial F}{\partial x} = & \sum_{\vec{a} \in A} \frac{-2(p_x - a_x)}{d(\vec{p}, \vec{a})^4} - \sum_{\vec{o} \in O} \frac{-2(p_x - o_x)}{d(\vec{p}, \vec{o})^4} \\ & - \sum_{\substack{\vec{e} \in E \\ \vec{e} \neq \vec{p}}} \frac{-2(p_x - e_x)}{d(\vec{p}, \vec{e})^4} \end{aligned} \quad (4)$$

$$\begin{aligned} \frac{\partial F}{\partial y} = & \sum_{\vec{a} \in A} \frac{-2(p_y - a_y)}{d(\vec{p}, \vec{a})^4} - \sum_{\vec{o} \in O} \frac{-2(p_y - o_y)}{d(\vec{p}, \vec{o})^4} \\ & - \sum_{\substack{\vec{e} \in E \\ \vec{e} \neq \vec{p}}} \frac{-2(p_y - e_y)}{d(\vec{p}, \vec{e})^4} \end{aligned} \quad (5)$$

The gradient determines the direction in which each evacuee moves. By normalizing the gradient vector and multiplying it by the evacuees speed (which was set to 2.62 ft/s, or 0.8 m/s [12]), the change in location of an evacuee during the current time-step is determined.

In this model, the contribution of a repulsive force on an evacuee decreases as the evacuee's distance from repulsor (i.e., obstacle or other evacuee) increases. However, in order to achieve more "realistic" evacuee movement, it was necessary allow evacuees to feel the repulsive effects of obstacles only if they were "close enough". (Otherwise, evacuees tend to move in unnecessarily wide arcs around obstacles.) To accomplish this, we first check to see if the $\vec{o}$ or $\vec{e}$ vectors are within a minimum distance of the current evacuee. If they are not, then we do not include the contribution of those particular repulsive vectors in our summation. A minimum repulsive distance of 0.1 ft was used in this simulation.

Finally, the model must account for the possibility of evacuees who are injured or killed when trying to escape. Thus, at each time-step, we check to see if the current evacuee is being overlapped by any other evacuee or obstacle by an amount greater than the evacuees radius. If so, the evacuee has been killed (by trampling or crushing) and, thus, may no longer move. However, the dead evacuees do not obstruct the movement of the other evacuees in any way, they no longer serve as repulsive forces, and they may be passed through by other evacuees. These simplifications were used because realistic parameters for such interactions between living and dead evacuees was unknown. Additionally, the EC was created to heavily penalize *any* casualties, as will be seen in the next section, so the interactions between living and dead evacuees should be minimal in the near-optimal solutions.

V. METHODOLOGY

A. Evaluation Function

Several issues needed to be balanced with regard to judging the fitness of a given set of exit locations. Most important is the simulation time needed for all evacuees to escape. The second issue is that the number of evacuees who were killed by crushing or trampling in their attempt to escape must carry a high penalty. The penalty should be considerably higher than the penalty for not escaping before the end of the simulation. This is because an evacuee who has been trampled to death has no chance of escaping, even if given extra time. Finally, the number of exit locations should be kept relatively low. Otherwise, the algorithm would likely

produce solutions where each evacuee has a door nearby. Since such a “solution” is physically unrealistic, a penalty for high numbers of doors was included. This penalty essentially represents a tradeoff of two seconds of escape time per exit (above the baseline of two exits). In other words, a given exit configuration with n exits ($n \geq 2$) and an escape time of t seconds would be equivalent to a configuration with $n + 1$ exits and an escape time of $t - 2$ seconds. (The previous example assumes that the casualties are unaffected by the additional exit.)

In order to calculate the fitness of a given configuration of exit locations, the simulation was run for a maximum of 20,000 milliseconds of simulation time. (This value was determined through preliminary experiments in which sub-optimal solutions for each layout were found which required less than 20,000 milliseconds for all evacuees to escape.) If all evacuees were able to exit safely in some time $t < 20000$, the fitness score for the configuration of exits was

$$t + 2000(\max(n, 2) - 2)$$

where n represents the number of doors in the configuration. Notice that for $n \leq 2$, the fitness becomes simply the time it takes for all evacuees to escape. Therefore the fitness will favor configurations with two doors or less.

If some evacuees were unable to escape, either because they were trampled to death or ran out of time, then the fitness score becomes

$$20000a + 30000d + 2000(\max(n, 2) - 2)$$

where a represents the number of evacuees who are still alive but have not escaped, and d represents the number of evacuees who were trampled to death.

B. Encoding Scheme

Each chromosome in the population was encoded as an array of N binary values. Here, N represents the maximum number of doors that may exist along the wall of the room. These elements represent whether a door should be placed at that location in the room and may be either 0 (no door) or 1 (door).

C. The Estimation of Distribution Algorithm

For the playroom layout, we used a non-elitist binary-coded EDA. This is because an individual may, by chance, be given a random set of initial locations of the evacuees for which it is ideally suited. In such a case, an elitist EDA would most likely keep this individual in the population for the remaining generations. However, a non-elitist EDA would force such an individual to produce well-suited offspring that can also fare well on different random configurations. Since the other two layouts (the lunchroom and classroom) did not contain random evacuee locations, we were able to use elitist EDAs where the best individual for each generation was allowed to survive. In all cases, the EDAs were constructed with a population size of 100.

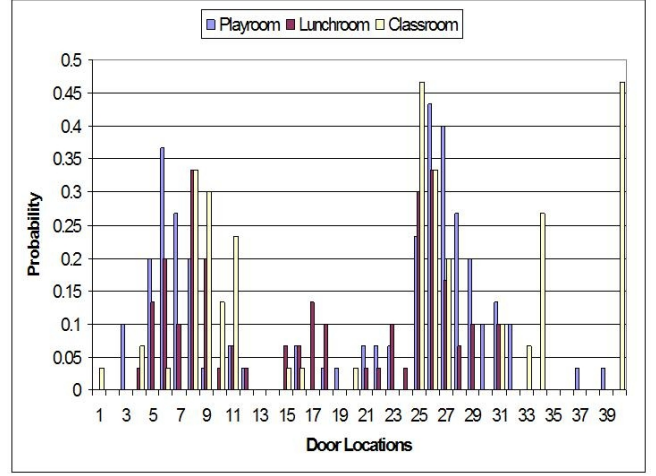


Fig. 4. Best Location Probabilities for All Layouts

D. Experimental Setup

For each layout, 30 different EDAs were created and allowed to evolve independently up to a maximum number of function evaluations (i.e., simulation runs of a given door configuration). For the playroom and classroom layouts, a maximum of 5000 function evaluations was used, while for the lunchroom layout, a maximum of 2500 evaluations was used. We used fewer function evaluations for the lunchroom layout because preliminary results revealed that the problem was “easier” than the others, most likely due to the symmetry of the layout.

VI. RESULTS

Each run of the EDA produced a best configuration of exits for the particular layout under consideration. Therefore, for each layout, 30 different best exit configurations were produced. These configurations were often unique solutions to the evacuation planning problem. Therefore, to identify similarities between solutions, we calculated the probability that each possible exit location was used across the 30 best solutions. These probabilities are presented in Fig. 4. In this graph, the x axis corresponds to the possible exit locations (40 possible exit locations for each layout), and the y axis represents the probability (out of the 30 runs) that a particular exit location actually contained a door. For instance, exit locations 25 and 40 had over a 45% chance of containing an exit for the classroom layout.

These statistics are useful, but the graph in Fig. 4 makes it difficult to truly understand the correspondence between the exit location and the particular layout for which it is likely. To provide illumination in this area, Fig. 5, 6, and 7 redisplay this information in the context of the room’s layout. In these figures, the probabilities associated with an exit location are coded as various shades of gray, where lighter colors represent higher probabilities. In each figure, probabilities greater than 20% (i.e., occurring in more than 6 of the 30 runs) are labeled with the actual value.

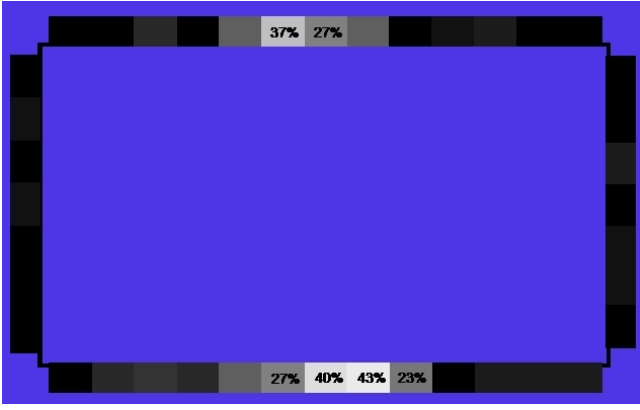


Fig. 5. Best Playroom Exit Location Probabilities

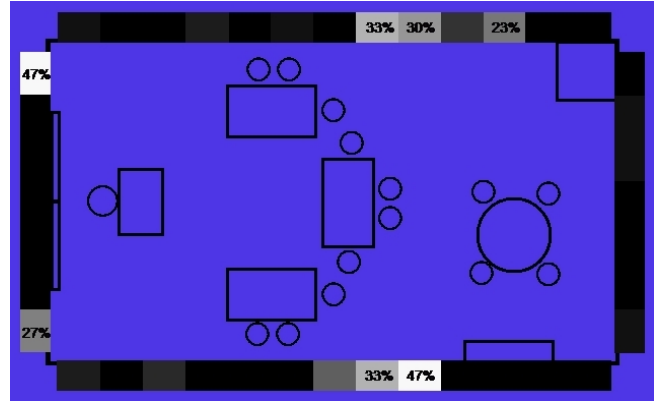


Fig. 7. Best Classroom Exit Location Probabilities

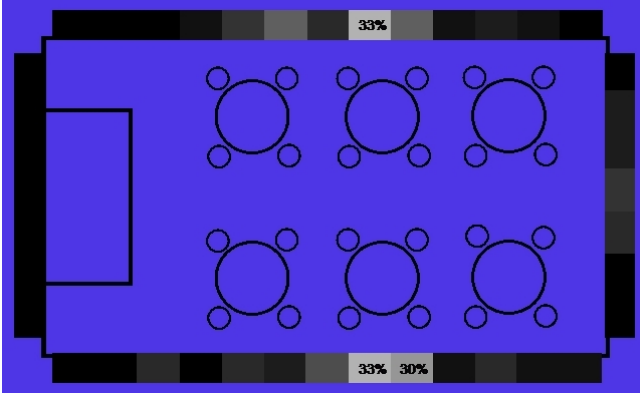


Fig. 6. Best Lunchroom Exit Location Probabilities

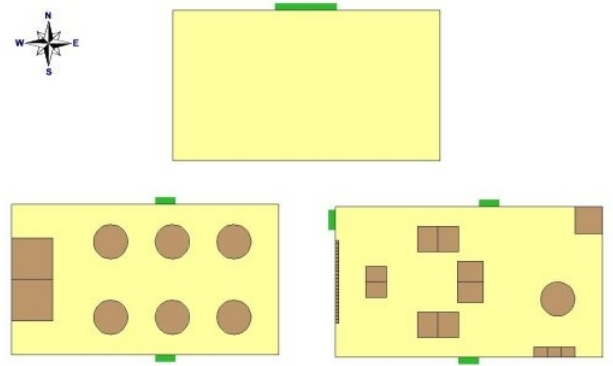


Fig. 8. Best Exit Placement for Each Layout

The best locations found for the playroom layout are shown in the upper layout of Fig. 8: three doors placed on the north wall. This arrangement of exit locations gave a total fitness of 12300. The average best fitness was 12810 with a standard deviation of 325.2. However, 53.3% of the best solutions found consisted of a total of four doors. A very near-optimal four-door arrangement (fitness of 12400) consisted of two doors each placed at the centers of the north and the south walls. The reason for this door arrangement may be due to the presence of symmetry inside the room.

The best locations found for the lunchroom layout are shown in the lower-left layout of Fig. 8: one door on both the north and south walls. This arrangement of exit locations gave a total fitness of 8300. The average best fitness of the solutions was 10286.7 with a standard deviation of 955.7. However, 56.7% of the results showed the presence of three doors. These exit locations include a door placed slightly east-of-center of the north wall and two doors placed slightly east-of-center of the south wall. The reason for the eastward shift in the door placement may be due to the presence of the stage in the west.

The best locations found for the classroom layout are shown in the lower-right layout of Fig. 8: one door placed near the center of the north and south walls, with a third door placed in the corner of the west wall. This arrangement of

exit locations gave a total fitness of 9000. The average best fitness for these solutions was 9460 with a standard deviation of 452.3. A total of 86.7% of the results consisted of three doors.

Table I summarizes the results from these experiments. The most striking feature of this table is that none of the best configurations experienced any casualties. The heavy penalty imposed on the fitness function for loss of life ensured that casualties would be non-existent. Additionally, we can see that the playroom and classroom layouts (which appear to be more complex) discovered solutions requiring more than two doors. This is somewhat surprising, given that such solutions were penalized. However, if a solution involving more than two doors provides a sufficient gain in speed of evacuation, its fitness will surpass that of solutions with fewer doors. It appears that the complexity of these layouts made it necessary and feasible to add more doors to aid in evacuation.

VII. CONCLUSIONS AND FUTURE WORK

It has been shown that evolutionary computation in the form of an Estimation of Distribution Algorithm could be used in evacuation planning to evolve locations of exits to optimize overall evacuation time. Although the technique is applied to a day-care setting, it has the potential to be

TABLE I
SUMMARY OF RESULTS

	Playroom	Lunchroom	Classroom
Best Fitness			
Average	12810	10286.7	9460
Standard Deviation	325.2	915.7	452.3
Function Evaluations			
Average	3183.3	1266.7	1536.7
Standard Deviation	898.7	235.4	390.0
Simulated Evacuation Time (ms)			
Average	9743.3	8753.3	7193.3
Standard Deviation	1009.8	1078.9	586.0
Casualties			
Average	0	0	0
Standard Deviation	0	0	0
Number of Doors			
Average	3.5	2.8	3.1
Standard Deviation	0.5	0.6	0.4

implemented in more complex layouts.

We believe that this area of study provides a great deal of promise and has been largely untapped. Future work from the evolutionary computation perspective should include the application of other types of ECs (e.g., genetic algorithms) to this problem in order to determine which types show better performance. From the evacuation planning perspective, experiments should be carried out to validate the results produced by this system with the guidelines set down in the Life Safety Code®. Finally, more complex layouts, perhaps involving more elaborate evacuation procedures, should be considered.

VIII. IN MEMORY OF

Our dear colleague, Dr. Brian Carnahan, passed on February 22, 2005 after a brief battle with a rare form of cancer. Dr.

Carnahan was an Associate Professor in the Department of Industrial and Systems Engineering at Auburn University. He came to Auburn University in August of 1999 after earning his Ph.D. from the University of Pittsburgh. Dr. Carnahan had also earned degrees from Pennsylvania State University and the University of Massachusetts. Brian leaves a wife, Paula, a daughter, and two twin boys. The authors truly miss his spirit, energy, dedication, creativity, warmth, and humor. We have lost a great friend and colleague.

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